

VENKATA RATNAM BHUMIREDDY | RESUME



- » **Status:** Staff, Analog IC Design Engineer at Vishay Siliconix, Cork, Ireland
- » **Contact:** Mail: venkata.ratnam.in17@gmail.com, Ph: +353 899 888 751
- » **Design Skills:** High Speed & Precision Analog, Digital (Verilog, P&R), Layout
- » **Tools:** Cadence, C, Skill, Verilog, Bash, Matlab, Perl, Ocean, mdl, Python
- » **Interests:** RF PLL, ADC, Serdes, PMIC, AI, ML, DevOps, Generative AI
- » **Activities:** Chess, Volleyball, Music, Reading

»»» Summary

- 14+ Years as Staff & Senior Mixed Signal ASIC Designer (10 years in Industry, 4.5 years at Tier 1 Research Centre)
- Working- Architecture and design of State-of-the-Art Temperature sensor at 0.25um
- Design of 2MSps Low Power 14Bit SAR ADC for PMIC at 180nm
- Designed and taped out 100MHz PMIC ASIC with 90% efficiency at 28nm for H2020 Project
- Part of team that designed 16 GHz All Digital PLL with 0.6ps rms Jitter for Serdes applications (2018)
- Past Industrial experience - design of SAR ADC, 32 Gbps Serdes Tx, Bandgaps, LDO, POR and Crystal Oscillator
- Worked on a variety of technology nodes from 180nm to 14nm CMOS
- 'Hands on experience' of layout, IO ring design and final chip assembly/tape-out
- Worked as Junior Research Fellow and Published IEEE ICCAS, Kuala Lumpur, 2013, pp. 1-4 - Completed online course work on Project Management, Product management, AI, ML and generative AI.
- Online course work on Project Management and Product management. Preparing for CAPM certification

»»» Work Experience

'22/06 - now	Staff, Analog IC Design Engineer	Vishay Siliconix, Cork, Ireland
	» Design of Optical Sensors and optical encoders	
'17/07 - '22/06	Senior Research Engineer	MCCI, Cork, Ireland
	» Design of Low Power SAR ADC for PMIC at 180nm (ongoing)	
	» Design/Layout of High speed Buck converter Power Path at 28nm ('19-'20)	
	» Design of High speed 16GHz RF Digital PLL at 28nm (0.6ps rms Jitter) ('17-'19)	
'15/02 - '17/07	Senior Design Engineer	GateLength Technology Pvt Ltd, Bangalore, India
	» Analysis and performance improvements of 16Gbps and 32Gbps SerDes Tx at 14nm	
	» Performance improvements in clock path and data path	
	» Simulation & characterization of macro level and top level in different modes of operation	
	» Supporting Layout engineers to improve performance	
'14/10 - '14/12	Consultant Engineer	Analog Semiconductors, Bangalore, India
	» Design of 1.6GHz CMOS Comparator for 100MSPS 14BIT SAR ADC at 65 nm	
	» Supporting top level integration of ADC	
	» Guiding layout engineers to achieve optimal design performance	
'12/10 - '14/12	Design Engineer	NXP Semiconductors India Pvt Ltd, Bangalore, India
	» Band gap Ref, POR, Pulse Generator and Ring Oscillator for switched regulator at 140nm.	
	» Design, simulation and characterization of Low Drop Out Regulator at 65nm	
	» Debugging and verification of 32MHz CMOS Crystal Oscillator at 140nm	
	» Performance improvements for client requirements	

'11/09 - '12/07	Junior Research Fellow	Dhirubhai Ambani Institute of ICT, Gandhinagar, India
	» Design of Low Power and High-Speed Comparator with Sub-32nm Double Gate MOSFET » Literature survey of "Comparators, sub-32nm technologies, and ADC".	
'08/11 - '10/07	Lecturer	Yuva Educational Society, Ongole, India
	» Teaching ECE subjects to prepare students for GATE examination for graduate studies » Prepare question papers and give counselling to improve the students performance.	

»»» Publications

2013	ICCAS	Kuala Lumpur
	» V. R. Bhumireddy et al., "Design of low power and high speed comparator with sub-32-nm Double Gate-MOSFET," 2013 IEEE International Conference on Circuits and Systems (ICCAS), Kuala Lumpur, 2013, pp. 1-4. doi: 10.1109/CircuitsAndSystems.2013.6671626 URL:(https://google.com)	

»»» Fellowships

2010 - 2011	Teaching Assistant	Dhirubhai Ambani Institute of ICT, Gandhinagar, India
2011 - 2012	Junior Research Fellow	Dhirubhai Ambani Institute of ICT, Gandhinagar, India

»»» Education

2010 - 2012	Master's Degree, VLSI	Dhirubhai Ambani Institute of ICT, Gandhinagar, India
	» Design of Low Power and High-Speed Comparator with Sub-32nm Double Gate MOSFET » Subjects: Analog CMOS IC Design, Digital CMOS IC Design, Signals and Systems » CGPA: 7.8/10	
2006 - 2008	Master of Science, Electronics	GVP PG College, Visakhapatnam, India
	» Subjects: Semiconductor Devices, Analog Electronics and Digital Electronics » Thesis: Industrial Timer and Controller using 8051 micro controller » Coding 8051 controller to display time and control the process for industrial applications » Percentage: 68.48%	
2003 - 2006	Bachelor of Science, Electronics	Sri Venkateswara University, Tirupati, India
	» Subjects: Mathematics, Physics, Electronics » Percentage: 72.30%	
2001 - 2003	12th Grade, Sciences	Board of Intermediate Education, India
	» Subjects: Mathematics, Physics, Chemistry » Percentage: 82.10%	
2000 - 2001	SSC, Sciences	Board of Secondary Education, India
	» Subjects: Mathematics, Physics, Chemistry » Percentage: 77.16%	

»»» References

- » References available on request